

Lecture 1 – Introduction to HAI

You should:

- Understand the definition of **interaction design** & interaction design goals
- Be able to work with and apply concepts of **affordances** and **signifiers**
+ how these work in genAI systems
- Understand the concept of **AI-as-design material**



Lecture 2 – Cognitive psychology and mental models in AI Design

- How perception / ‘types of thinking’ work + why is this key for design
- Conceptual vs represented vs mental models
 - How these relate to gulfs of evaluation / execution
 - How these concepts link to affordances & signifiers
- Types of distributed cognition (cognitive / computational offloading, annotating...)
 - ... and in particular the ability to see a designed system & analyse it in these terms.
- Principles of the Computers as Social Actors theories



Lecture 3 – Interface and Interaction Patterns for Human-AI Systems

- Understand how usability goals & heuristics relate to the content of Lectures 1-2
- Remember & be able to use the 6 usability goals
- Remember & be able to use the 10 heuristics
(... + understand what a heuristic is as a concept)
- Be able to compare and contrast these concepts to emerging frameworks for human-AI collaboration

Lecture 4 – User needs & problem definition

- Principles of human-centred design & how this is represented in the lifecycle / design models
- Differences between needs & requirements
 - + how to gather data & develop each (methods for interviews, observations ...)
- What are empathy maps & how to use them (see LGT)
- Human participation in AI models development



Lecture 5 – Prototyping

- Definition and function of prototypes ...
... what are they? What role do they play in design? How do these related to assumption testing?
- Types of prototypes & their respective pros / cons
- Prototyping methods and approaches



Lecture 7 – Automation, agency & fairness

- Understand and be able to apply the concepts of Control vs Automation
... including the ability to analyse design systems in these terms.
- Definition of fairness and prior challenges with AI systems
- Definitions of harms, types of harms, and their application
... including the ability to analyse an existing system to identify possible harms
- Concepts of Transparency vs Explainability



Lecture 8 – Environmental Sustainability

- Understand the potential sources of environmental impact from AI systems
- Broad understanding of how these calculations are made

Lecture 9 – Evaluation



- Purpose of evaluation – pros / cons of different methods
- Understand differences between
 - precision, generalizability, realism (+ concrete/abstract – obtrusive /unobtrusive)
 - approaches involving / not involving users
- Specific methods in detail -- heuristic evaluation + think aloud
- Theory and methods for experiments, including ability to apply key terms:
 - Correlation vs causation; hypothesis, independent, dependent, uncontrolled variables
 - Concepts of reliability and validity (and their different subtypes)
 - Between / within subjects designs

Lecture 10 – red teaming and field studies

- Understand and be able to apply the concept of **appropriation** of technology
- Technology probes – definition, role in HAI design, differences to prototypes
- Red teaming
 - Definition, why is it needed with genAI systems
 - How red-teaming relates to harms
 - Types of potential attacks
 - Potential defences (including spotlighting / constitutional classifiers)

